

Properties of equality and congruence



1

a $b + c$

b bc

c AB

2 The multiplication property of equality.

3

a The symmetric property of equality.

b The addition property of equality.

c The transitive property of congruence.

4

a No

b No

c Yes

d No

5

a Yes

b Yes

c No

d Yes

6

a No

b No

c No

d Yes

7

a The transitive property of equality.

b The multiplication property of equality.

c The addition property of equality.

8

a No

b Yes

c No

d No

e Yes

f No



Additional questions

9

Property of Equality	Example
Reflexive property of equality	$a = a$
Symmetric property of equality	If $a = b$, then $b = a$.
Transitive property of equality	If $a = b$ and $b = c$, then $a = c$.
Addition property of equality	If $a = b$, then $a + c = b + c$.
Subtraction property of equality	If $a = b$, then $a - c = b - c$.
Multiplication property of equality	If $a = b$, then $a \times c = b \times c$.
Division property of equality	If $a = b$ and $c \neq 0$, then $a \div c = b \div c$.
Substitution property of equality	If $a = b$, then b may be substituted for a in any expression containing a .

10

- a Addition property of equality
- b Converse of multiplication property
- c Converse of addition property of equality
- d Converse of subtraction property of equality

11

- a Converse of transitive property of equality
- b Substitution property of equality